Finding Out What Went Wrong

Two different kinds of wrong

An error

The program stops or refuses to run. The console names the line. A missing bracket, a misspelled function. **Loud, and easy.**

A bug

The program runs perfectly and does the wrong thing. Nothing is reported. A shape drawn too late, an offset from the wrong number. **Quiet, and hard.**

> Almost nothing that went wrong in this unit produced an error message. The computer did exactly what was written every time — which was not what anyone meant.

When nothing is reported — four questions, in order

ASK	HOW
Is this line running at all?	`console.log("got here")`
Is this variable changing?	`console.log(progress)`
Is it where I think it is?	inside `draw()`, or outside?
Is it spelled exactly right?	`keyPressed, not keypressed`

> **Change one thing, then run it.** Two changes and you will not know which one worked — and you will not be able to do it again next time.

The three bugs you met today

SYMPTOM	WHAT IS ACTUALLY WRONG	WHICH QUESTION CATCHES IT
The bar has no fill	A fixed `0` in the width slot instead of the variable	Look at the shape call — is the variable where you think it is?
Pressing a key does nothing	The handler is nested *inside* `draw()`	Is it where I think it is? p5 looks for handlers at the top level
Nothing grows at all	`let` is inside `draw()`, so the value resets every frame	Is the variable changing? Log it and see the same number print over and over

The pattern behind today's sketch

A flag controls growth

```
let loading = 0;                          // 0 = stopped, 1 = growing

progress = progress + (2 * loading);      // adds 2 when loading is 1,
                                          // adds 0 when loading is 0

function keyPressed()   { loading = 1; }  // start
function mousePressed() { loading = 0; }  // stop
```

> **Multiplying by zero makes the addition do nothing.** That is how we stop growth without `if`. A variable that is either 1 or 0 is called a **flag**.

Mapping a value to a percentage

```
progress = progress + (2 * loading);
percent  = round(progress * 100 / 340);   // 340 is the track width
```

`progress` is the width in *pixels*. `percent` is the same value expressed 0 to 100. They are two different quantities, computed from the same growth.

Useful extras — not core topics

TOOL	WHAT IT DOES
`round(x)`	Rounds to the nearest whole number. `round(42.7)` gives `43`. Met it today because the percentage had a long decimal.

Symptoms you have already met

WHAT YOU SEE	USUALLY MEANS
Nothing moves at all	`let` is inside `draw()`, so it resets every frame
A shape is missing	something drawn after it is covering it
A shape vanished completely	it is above `background()`, erased every frame
Text is invisible	no `fill()` before `text()`
Everything down and right by half its size	`rectMode(CENTER)` missing
One part left behind while the rest moves	that part still has a fixed number instead of an offset
A handler does nothing, no error	misspelled, or written inside `draw()`
A number prints with a long decimal	wrap the arithmetic in `round()`

On the AP exam

Written Response 2(b) asks you to describe an error in your program and how you identified it. The answer that scores is a **process**, not a lucky guess:

> *"The loading bar was not growing. I added `console.log(progress)` and saw it printing 0 over and over. That told me the value was resetting every frame, so the `let` was inside `draw()`. I moved it above `setup()` and the bar started growing."*

What we cannot do yet. The loading bar does not stop at 100%. We cannot check *"has it gone past the end?"* without `if`. That arrives in Unit 2.

Unit 1 covered CED topics 3.1, 3.3, 3.4 and 1.4 — variables and assignment, developing algorithms, strings, and debugging. All inside Big Idea 3, which is 30–35% of the multiple-choice exam.